ON A CONNECTED SCHEME A BOUNDED BELOW CONSTRUCTIBLE COMPLEX WITH LOCALLY CONSTANT COHOMOLOGY IS CONSTANT ON AN ETALE COVERING OVER FINITE PROJECTIVE MODULES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the trivialising covering, the grouping into disjoint open pieces by isomorphism class of the local model with the reason for each step, the connectedness conclusion, and the perfectness criterion making the model finite projective.

Editorial note. The label lemma-connected-ctf-locally-constant names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The constructible finite type derived category with locally constant cohomology. Same chapter, the definition of the full subcategory of complexes with constructible cohomology sheaves and the lemma identifying the constants for a locally constant sheaf of Lambda-modules. Supplies the hypothesis on K and the local constancy description used at the start of the proof.
- Perfectness of the local models. More on Algebra, the lemma characterising perfect complexes over Noetherian rings through the finite type and boundedness properties. Supplies that the module complexes M_i^bullet are perfect, hence replaceable by finite complexes of finite projective modules.

Source reference. Original lemma 0658: sources/stacks-chow/tags.txt, source lines 7892–7892. Prerequisite pointer invoked at proof line(s) 13875; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 094G: sources/stacks-chow/tags.txt, source lines 11549–11549. Prerequisite pointer invoked at proof line(s) 13863; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a connected scheme. Let Λ be a Noetherian ring. Let $K \in D_{ctf}(X_{étale}, \Lambda)$ have locally constant cohomology sheaves. Then there exists a finite complex of finite projective Λ-modules $M^\bullet$ and an étale covering $\{U_i \to X\}$ such that $K|_{U_i} \cong \underline{M^\bullet}|_{U_i}$ in $D(U_{i,étale}, \Lambda)$.*

Proof. Choose an étale covering $\{U_i \to X\}$ such that $K|_{U_i}$ is constant, say $K|_{U_i} \cong \underline{M_i^\bullet}_{U_i}$ for some finite complex of finite Λ-modules $M_i^\bullet$. See Cohomology on Sites, Lemma 094G. Observe that $U_i \times_X U_j$ is empty if $M_i^\bullet$ is not isomorphic to $M_j^\bullet$ in $D(\Lambda)$. For each complex of Λ-modules $M^\bullet$ let $I_{M^\bullet} = \{i \in I \mid M_i^\bullet \cong M^\bullet \text{ in } D(\Lambda)\}$. As étale morphisms are open we see that $U_{M^\bullet} = \bigcup_{i \in I_{M^\bullet}} \mathrm{Im}(U_i \to X)$ is an open subset of X. Then $X = \coprod U_{M^\bullet}$ is a disjoint open covering of X. As X is connected only one $U_{M^\bullet}$ is nonempty. As K is in $D_{ctf}(X_{étale}, \Lambda)$ we see that $M^\bullet$ is a perfect complex of Λ-modules, see More on Algebra, Lemma 0658. Hence we may assume $M^\bullet$ is a finite complex of finite projective Λ-modules. $\square$